BUILD YOUR PERSONAL CYBERSECURITY LAB

Internship Task-2 Lab Report

Maincrafts Technology

Field	Details
Intern Name	Eavam Margamwar
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Organization	Maincrafts Technology
Task	Task-2: Personal Cybersecurity Lab Setup
Attacker Machine	Kali Linux 2025.4 (VMware)
Target Application	OWASP Juice Shop (Docker)

1. Objective

The objective of this task is to build a safe and isolated cybersecurity practice environment on a personal computer using virtualization. This hands-on lab serves as the foundation for future tasks including vulnerability assessment, web application testing, exploitation practice, and incident response.

The lab environment consists of:

- 1 Attacker machine — Kali Linux 2025.4 (VMware)
- 1 Vulnerable target application — OWASP Juice Shop (running via Docker)
- Proper network segmentation using NAT and Host-Only adapters

2. System Specifications

Requirement	My System
Operating System	Windows (Host)
Virtualization Software	VMware Workstation
RAM Allocated to VM	3-4 GB
vCPUs	2
Disk Space Used	~40 GB
Kali Linux Version	2025.4 (VMware AMD64)
Target App	OWASP Juice Shop v17+ (Docker)

3. Network Architecture

Two network adapters were configured on the Kali Linux VM to separate internet traffic from lab traffic:

Interface	Configuration
eth0 (Adapter 1)	NAT — 192.168.61.129/24 — Internet access for updates
eth1 (Adapter 2)	Host-Only — 192.168.85.128/24 — Isolated lab traffic
docker0	172.17.0.1/16 — Docker bridge for Juice Shop container

This architecture ensures that all attack traffic stays within the isolated Host-Only network and Docker bridge, with no risk to the real network or host system.

4. Step-by-Step Setup

Step 1 — Install VMware and Configure Networks

VMware Workstation was installed on the host machine. Two virtual networks were configured:

- VMnet8 (NAT) — provides internet access to VMs
- VMnet1 (Host-Only) — provides isolated lab network

Step 2 — Deploy Kali Linux (Attacker VM)

The official Kali Linux VMware image was downloaded from kali.org and imported into VMware. Two network adapters were assigned: NAT (eth0) and Host-Only (eth1). RAM was set to 4GB and CPUs to 2.

System update command used:

```
sudo apt clean && echo "deb https://kali.download/kali kali-rolling main non-free
contrib" | sudo tee /etc/apt/sources.list && sudo apt update
```

Step 3 — Deploy OWASP Juice Shop (Target)

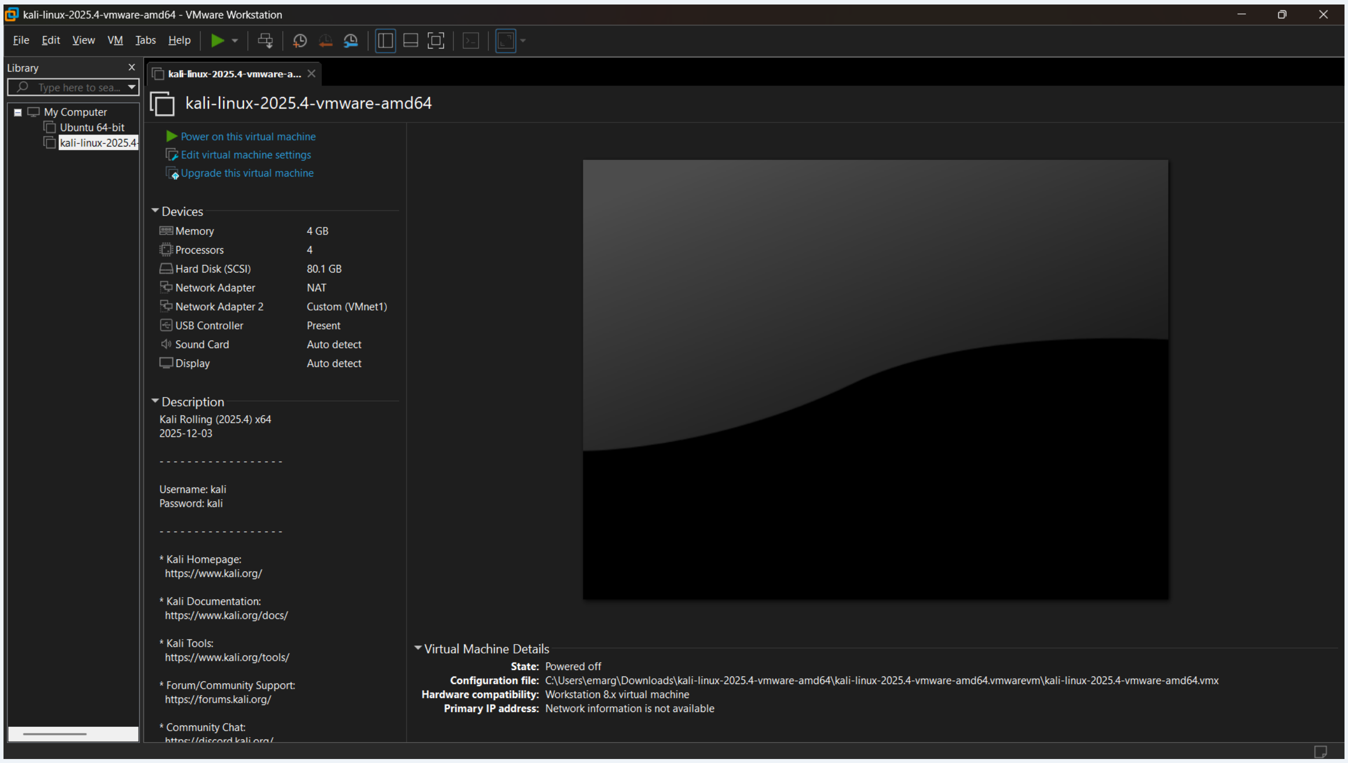
Docker was installed and OWASP Juice Shop was deployed as a container on port 3000:

```
sudo apt install -y docker.io && sudo systemctl enable --now docker
sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
```

5. Evidence and Screenshots

5.1 VM List

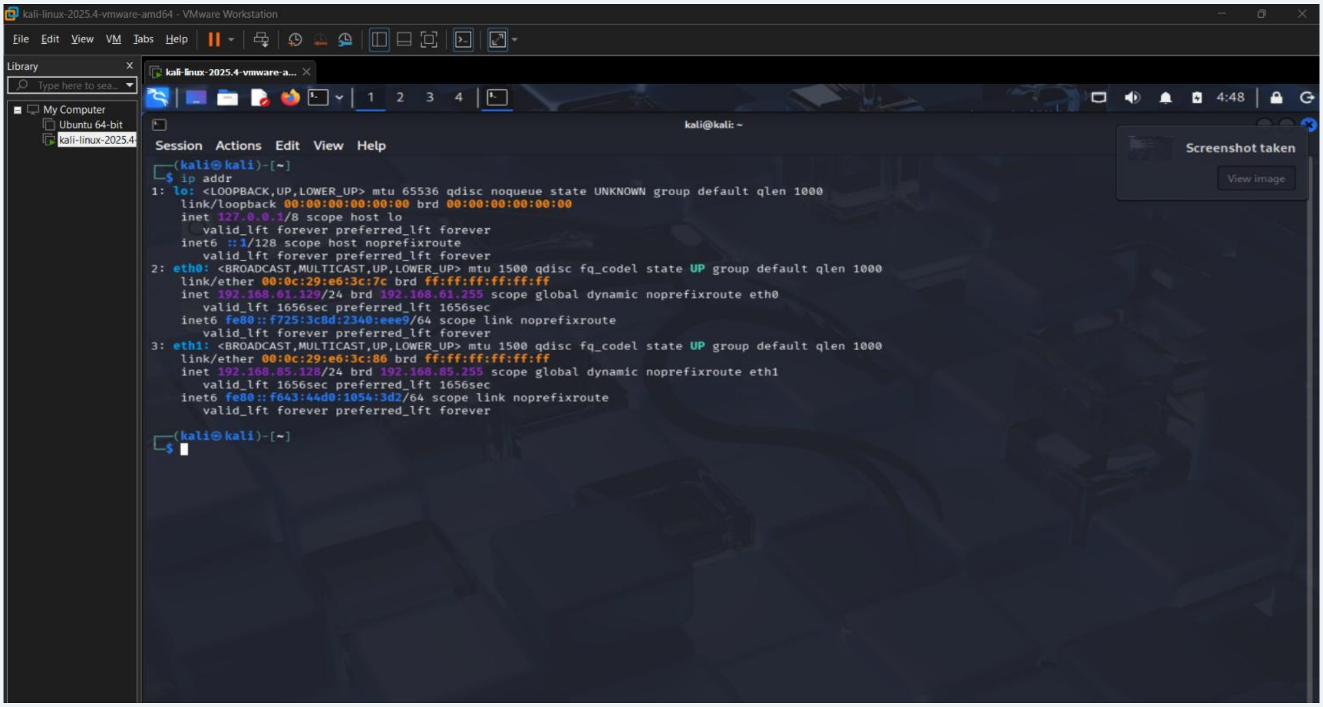
The following screenshot shows the Kali Linux VM running in VMware Workstation.



VMware Workstation — Kali Linux VM running

5.2 IP Configuration

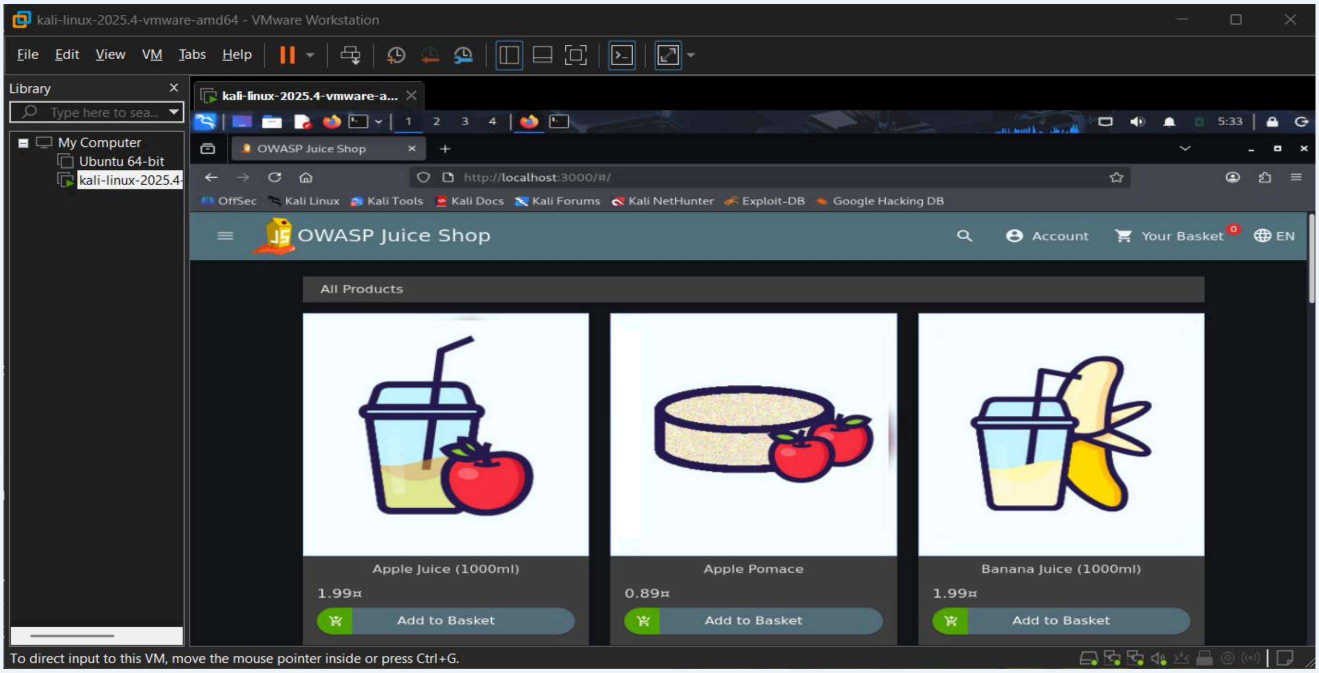
The ip addr command was run inside Kali Linux to confirm both network adapters were active. eth0 (NAT) received 192.168.61.129 and eth1 (Host-Only) received 192.168.85.128.



Terminal — ip addr output showing eth0 and eth1

5.3 Target Application — OWASP Juice Shop

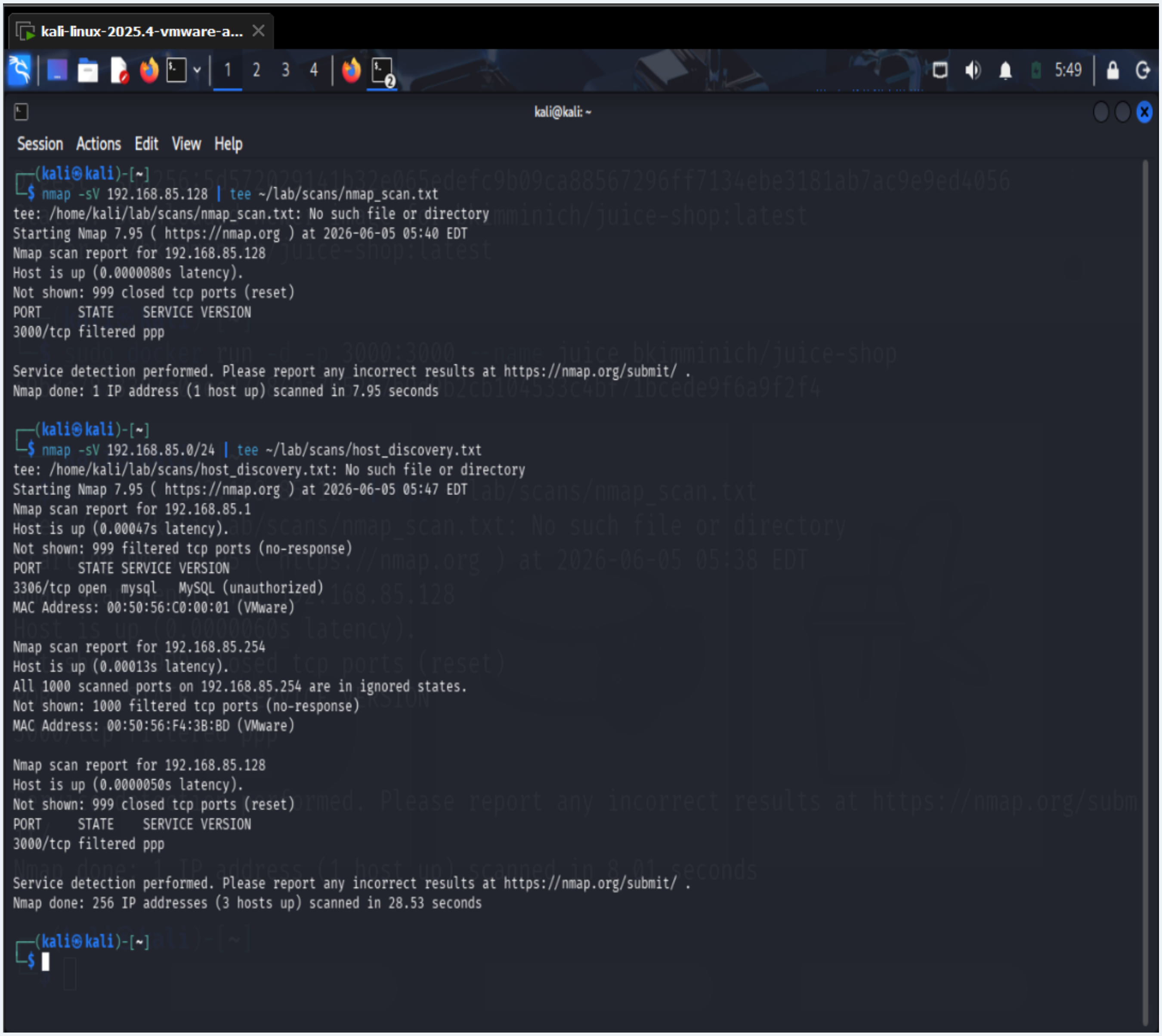
OWASP Juice Shop was successfully deployed via Docker and accessed in the Firefox browser inside Kali at http://localhost:3000.



Firefox — OWASP Juice Shop welcome page at localhost:3000

5.4 Nmap Scan

Nmap was used to perform host discovery and service detection on the Host-Only subnet and localhost. The scan confirmed Juice Shop running on port 3000 and identified other hosts on the 192.168.85.0/24 network.

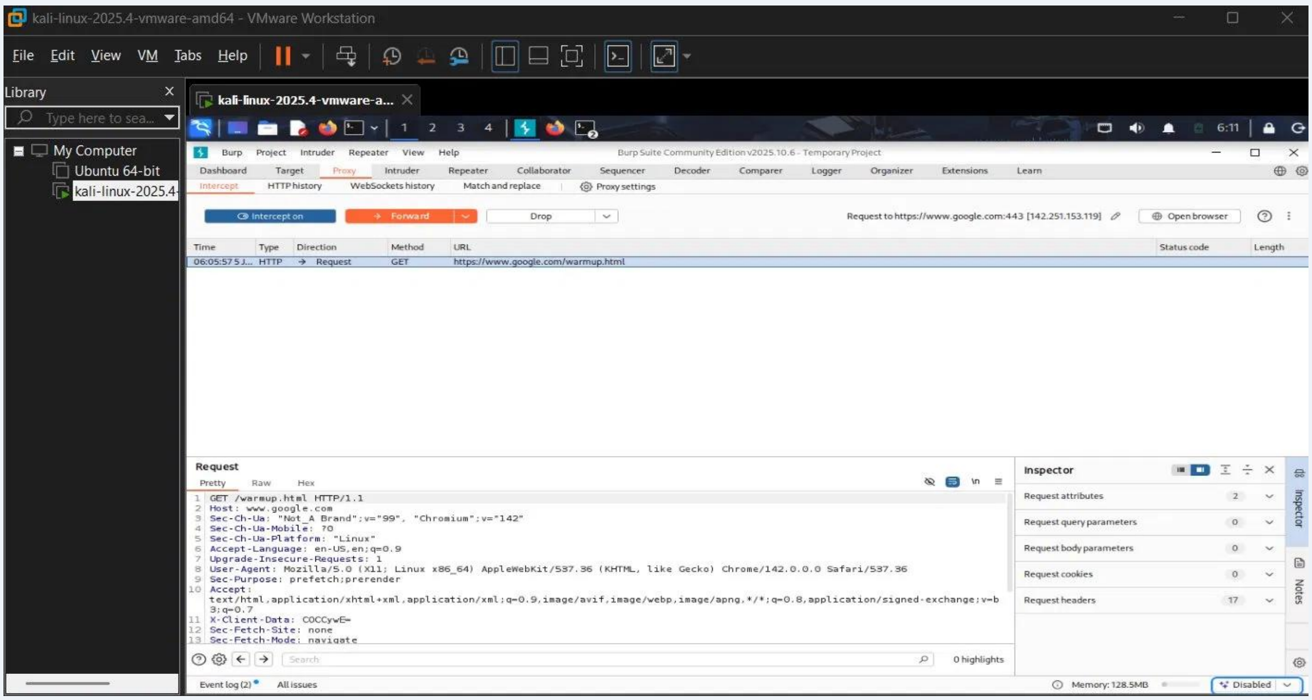


Terminal — Nmap scan output (nmap -sV localhost and subnet scan)

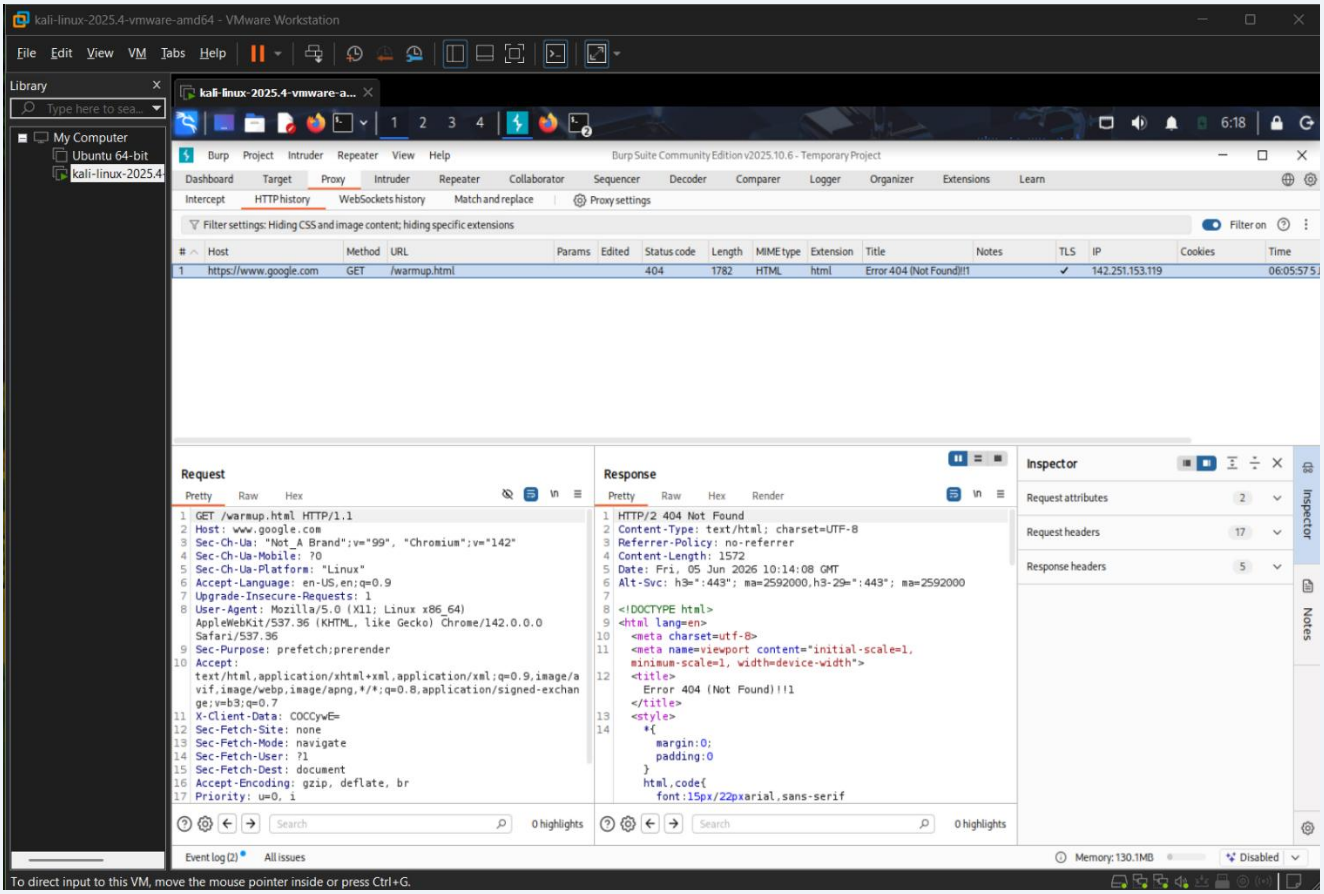
Scan Detail	Result
Command used	nmap -sV 192.168.85.0/24
Hosts discovered	3 (192.168.85.1, .128, .254)
Notable open port	3306/tcp — MySQL on 192.168.85.1 (VMware gateway)
Juice Shop port	3000/tcp (docker0 bridge, accessible via localhost)

5.5 Burp Suite — Traffic Interception

Burp Suite Community Edition v2025.10.6 was launched in Kali. Firefox was configured to route traffic through Burp proxy at 127.0.0.1:8080. HTTP requests to Juice Shop were successfully intercepted and analysed.



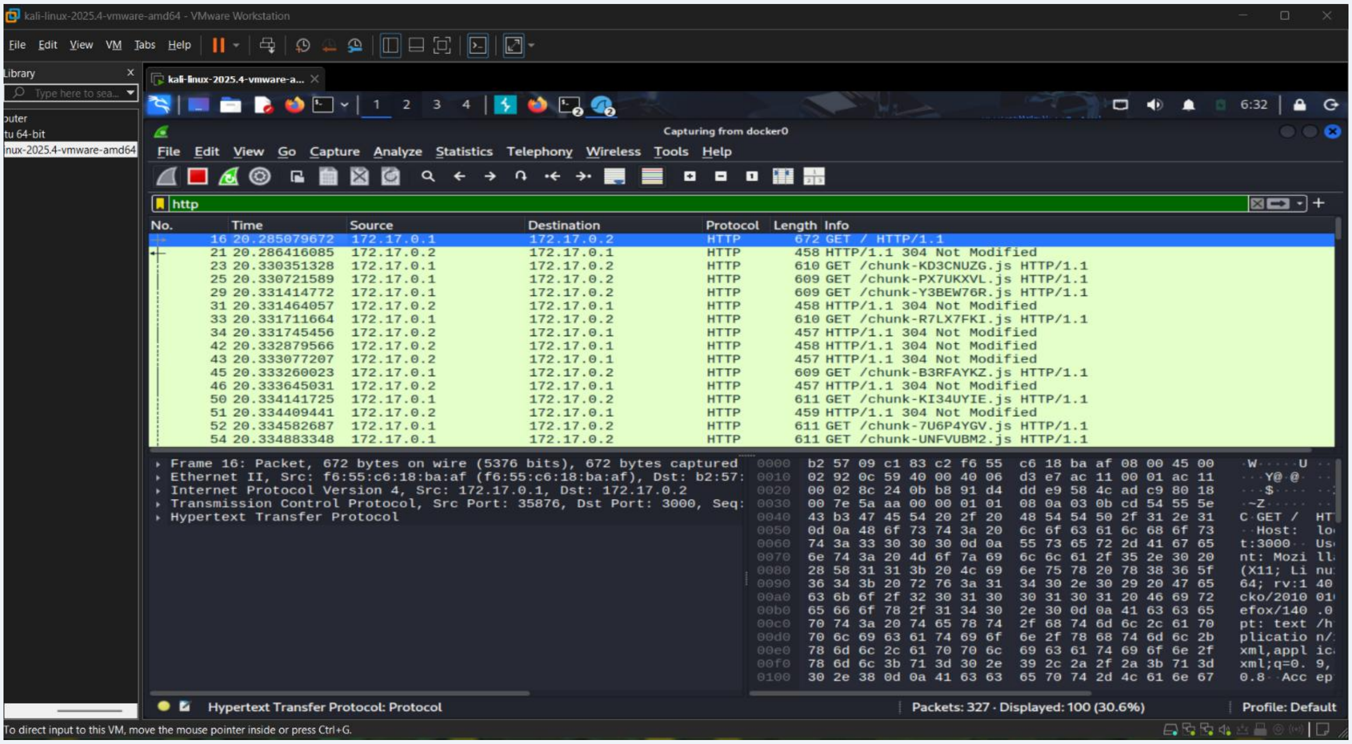
Burp Suite — Intercept tab showing captured HTTP request



Burp Suite — HTTP History tab showing request/response details

5.6 Wireshark — Packet Capture

Wireshark 4.6.0 was used to capture live HTTP traffic on the docker0 interface. HTTP packets between the host (172.17.0.1) and the Juice Shop container (172.17.0.2) were clearly visible, including GET requests for the application's JavaScript files.



Wireshark — docker0 interface capturing HTTP packets to/from Juice Shop

Capture Detail	Result
Interface captured	docker0
Display filter	http
Source IP	172.17.0.1 (Kali host)
Destination IP	172.17.0.2 (Juice Shop container)
Packets captured	327 total, 100 HTTP displayed

6. Deliverables Checklist

Deliverable	Screenshot	Status
Kali Linux VM deployed and running	VM list screenshot	**Done**
IP configuration confirmed (eth0 + eth1)	ip addr screenshot	**Done**
OWASP Juice Shop deployed via Docker	Browser screenshot	**Done**
Nmap host discovery and port scan	Terminal screenshot	**Done**
Burp Suite traffic interception	Burp intercept screenshot	**Done**
Burp Suite HTTP history	HTTP history screenshot	**Done**
Wireshark packet capture on docker0	Wireshark screenshot	**Done**

7. Tools Used

Tool	Purpose
VMware Workstation	Virtualization platform to run Kali Linux VM
Kali Linux 2025.4	Attacker machine with pre-installed security tools
Docker	Container engine to deploy OWASP Juice Shop
OWASP Juice Shop	Intentionally vulnerable web application (target)
Nmap 7.95	Network scanning and host/service discovery
Burp Suite Community v2025.10.6	HTTP proxy for intercepting and analysing web traffic
Wireshark 4.6.0	Network packet capture and protocol analysis

8. Key Learnings

- Understood how virtualization creates isolated environments for safe security testing
- Configured dual-network adapters (NAT + Host-Only) to separate internet and lab traffic
- Deployed a vulnerable web application using Docker containers
- Used Nmap to discover hosts and open services on a local network
- Intercepted and analysed HTTP traffic using Burp Suite proxy
- Captured and filtered network packets using Wireshark on the docker0 interface
- Troubleshot Kali repository mirror failures and switched to the kali.download mirror

9. Conclusion

The personal cybersecurity lab has been successfully built and validated. The Kali Linux attacker machine is running with proper dual-network configuration, OWASP Juice Shop is deployed and accessible as the target application, and all required tools (Nmap, Burp Suite, Wireshark) have been verified to work correctly.

This isolated lab environment provides a safe foundation for all future cybersecurity tasks including vulnerability assessment, web application penetration testing, exploitation practice, and incident response — without any risk to real systems or networks.